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Inventor(s)	Park; Sung Il et al.

Semiconductor memory devices and methods for manufacturing the same

Abstract

A semiconductor memory device and a method for manufacturing the same. The semiconductor memory device may include a substrate, a first lower wire pattern and a first upper wire pattern stacked on the substrate, and spaced apart from each other; a second lower wire pattern and a second upper wire pattern stacked on the substrate, spaced apart from each other, and spaced apart from the first lower and upper wire patterns; a first gate line surrounding the first lower wire pattern and the first upper wire pattern; a second gate line surrounding the second lower wire pattern and the second upper wire pattern and spaced apart from the first gate line; a first lower source/drain area; a first upper source/drain area; and a first overlapping contact that electrically connects the first lower source/drain area, the first upper source/drain area and the second gate line to each other.

Inventors: **Park; Sung Il (Suwon-si, KR), Park; Jae Hyun (Hwaseong-si, KR), Kim; Min Gyu (Hwaseong-si, KR), Choi; Do Young (Hwaseong-si, KR), Ha; Dae Won (Seoul, KR)**

Applicant: **Samsung Electronics Co., Ltd.** (Suwon-si, KR)

Family ID: **84541260**

Assignee: **SAMSUNG ELECTRONICS CO., LTD. (Suwon-si, KR)**

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Primary Examiner: Nadav; Ori

Attorney, Agent or Firm: Myers Bigel, P.A.

Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION

(1) This application claims priority from Korean Patent Application No. 10-2021-0082023 filed on Jun. 24, 2021, in the Korean Intellectual Property Office, and all the benefits accruing therefrom under 35 U.S.C. § 119, and the entire contents of the above-identified application are herein incorporated by reference.

TECHNICAL FIELD

(2) Aspects of the present disclosure relate to semiconductor memory devices and methods for manufacturing the same.

BACKGROUND

(3) One way semiconductor memory devices may be classified is into volatile memory devices and non-volatile memory devices. A volatile memory device is a memory device in which stored data therein is removed or lost when power supply to the memory device is removed or turned off, and examples may include SRAM (Static RAM), DRAM (Dynamic RAM), and SDRAM (Synchronous DRAM). A non-volatile memory device is a memory device that retains stored data therein even when power supply to the memory device is lost or removed, and examples may include ROM (Read Only Memory), PROM (Programmable ROM), EPROM (Electrically Programmable ROM), EEPROM (Electrically Erasable and Programmable ROM), flash memory devices, resistance memory devices such as PRAM (Phase-change RAM), FRAM (Ferroelectric RAM), RRAM (Resistive RAM), and the like.

(4) While DRAM uses a capacitor to store data therein, SRAM may store data therein using a latch. SRAM has lower integration density compared to DRAM, but has advantages in that a peripheral circuit thereof is simple, SRAM operates at high speed with low power, and does not need to periodically refresh stored information.

(5) As integration levels of semiconductor memory devices are increasing, individual circuit patterns are becoming more miniaturized in order to implement a larger number of semiconductor memory devices in the same area. To this end, semiconductor memory devices that use multi-gate transistors are being studied.

SUMMARY

(6) Some aspects of the present disclosure provide semiconductor memory devices with improved integration density and reduced process difficulty.

(7) Some aspects of the present disclosure provide methods for manufacturing semiconductor memory devices with improved integration density and reduced process difficulty.

(8) The present disclosure is not limited to the aspects provided above or to those explicitly stated herein. Other aspects and purposes of the present disclosure that are not mentioned herein may be understood based on following descriptions, and may be more clearly understood based on examples of embodiments of the inventive concepts provided by the present disclosure. Further, it will be easily understood that some aspects, purposes, and advantages according to the present disclosure may be realized using means shown in the claims and combinations thereof.

(9) According to some aspects of the present inventive concepts, there is provided a semiconductor memory device comprising a substrate, a first lower wire pattern and a first upper wire pattern sequentially stacked on the substrate, and spaced apart from each other, each extending in a first direction, a second lower wire pattern and a second upper wire pattern sequentially stacked on the substrate, and spaced apart from each other, each extending in the first direction, the second lower wire pattern and the second upper wire pattern spaced apart from the first lower wire pattern and the first upper wire pattern in a second direction that intersects the first direction, a first gate line extending in the second direction, and surrounding the first lower wire pattern and the first upper wire pattern, a second gate line extending in the second direction, and surrounding the second lower wire pattern and the second upper wire pattern, the second gate line spaced apart from the first gate line in the second direction, a first lower source/drain area having a first conductivity

type, on one side surface of the first gate line, and connected to the first lower wire pattern, a first upper source/drain area having a second conductivity type different from the first conductivity type, on one side surface of the first gate line, and connected to the first upper wire pattern, and a first overlapping contact that electrically connects the first lower source/drain area, the first upper source/drain area and the second gate line to each other, wherein the first overlapping contact at least partially vertically overlaps the first gate line, wherein the first gate line includes a first gate electrode and a recess capping pattern, wherein the recess capping pattern covers a top surface of the first gate electrode that overlaps the first overlapping contact, wherein the second gate line includes a second gate electrode and a gate capping pattern, wherein the gate capping pattern covers a top surface of the second gate electrode, and wherein a vertical level of a bottom surface of the recess capping pattern is lower than a vertical level of a bottom surface of the gate capping pattern.

(10) According to some aspects of the present inventive concepts, there is provided a semiconductor memory device comprising a substrate, a first lower wire pattern and a first upper wire pattern sequentially stacked on the substrate, and spaced apart from each other, each extending in a first direction, a second lower wire pattern and a second upper wire pattern sequentially stacked on the substrate, and spaced apart from each other, each extending in the first direction, the second lower wire pattern and the second upper wire pattern spaced apart from the first lower wire pattern and the first upper wire pattern in a second direction that intersects the first direction, a first gate line extending in the second direction, and surrounding the first lower wire pattern and the first upper wire pattern, a second gate line extending in the second direction, and surrounding the second lower wire pattern and the second upper wire pattern, wherein the second gate line is spaced apart from the first gate line in the second direction, a first lower source/drain area having a first conductivity type, on one side of the first gate line, and connected to the first lower wire pattern, a first upper source/drain area having a second conductivity type different from the first conductivity type, on one side of the first gate line, and connected to the first upper wire pattern, and a common contact extending in a third direction that intersects a top surface of the substrate, wherein the common contact is connected to the first lower source/drain area and the first upper source/drain area, and an overlapping contact that electrically connects the common contact and the second gate line to each other, the overlapping contact at least partially overlapping the first gate line.

(11) According to some aspects of the present inventive concepts, there is provided a semiconductor memory device comprising a substrate, a first lower wire pattern and a first upper wire pattern sequentially stacked on the substrate, and spaced apart from each other, each extending in a first direction, a second lower wire pattern and a second upper wire pattern sequentially stacked on the substrate, and spaced apart from each other, each extending in the first direction, the second lower wire pattern and the second upper wire pattern spaced apart from the first lower wire pattern and the first upper wire pattern in a second direction that intersects the first direction, a first gate line extending in the second direction, and surrounding the first lower wire pattern and the first upper wire pattern, a second gate line extending in the second direction, and surrounding the second lower wire pattern and the second upper wire pattern, wherein the second gate line is spaced apart from the first gate line in the second direction, a third gate line extending in the second direction, and surrounding the first lower wire pattern and the first upper wire pattern, and spaced apart from the first gate line in the first direction, a fourth gate line extending in the second direction, and surrounding the second lower wire pattern and the second upper wire pattern, and spaced apart from the third gate line in the second direction, a first lower source/drain area having a first conductivity type, between the first gate line and the third gate line, and connected to the first lower wire pattern, a first upper source/drain area having a second conductivity type different from the first conductivity type, between the first gate line and the third gate line, and connected to the first upper wire pattern, and a first overlapping contact that electrically connects the first lower source/drain area, the first upper source/drain area, and the second gate line to each other, wherein

the first overlapping contact at least partially overlaps the first gate line, wherein each of the first to fourth gate lines includes a gate electrode and a gate capping pattern that covers a top surface of the gate electrode, wherein the first gate line further includes a first recess capping pattern that covers a top surface of the gate electrode of the first gate line overlapping the first overlapping contact, and wherein a vertical level of a bottom surface of the first recess capping pattern is lower than a vertical level of a bottom surface of the gate capping pattern.

Description

BRIEF DESCRIPTION OF DRAWINGS

(1) The above and other aspects and features of the present disclosure will become more apparent by describing in detail some examples of embodiments thereof with reference to the attached drawings, in which:

(2) FIG. **1** is a circuit diagram for illustrating a semiconductor memory device according to some embodiments;

(3) FIG. **2** is an exemplary layout diagram for illustrating the semiconductor memory device of FIG. **1**;

(4) FIG. **3** is a cross-sectional view taken along a line A-A of FIG. **2**;

(5) FIG. **4** is a cross-sectional view taken along a line B-B in FIG. **2**;

(6) FIG. **5** is a cross-sectional view taken along a line C-C in FIG. **2**;

(7) FIGS. **6** to **8** are various cross-sectional views for illustrating a semiconductor memory device according to some embodiments;

(8) FIG. **9** is an exemplary layout diagram for illustrating a semiconductor memory device according to some embodiments;

(9) FIG. **10** is an exemplary layout diagram for illustrating a semiconductor memory device according to some embodiments;

(10) FIG. **11** is a cross-sectional view taken along a line D-D of FIG. **10**;

(11) FIG. **12** is a cross-sectional view taken along a line E-E of FIG. **10**;

(12) FIG. **13** to FIG. **25** are diagrams showing intermediate steps for illustrating operations of a method for manufacturing a semiconductor memory device according to some embodiments;

(13) FIG. **26** is a diagram showing an intermediate step for illustrating operations of a method for manufacturing a semiconductor memory device according to some embodiments; and

(14) FIG. **27** is a diagram showing an intermediate step for illustrating operations of a method for manufacturing a semiconductor memory device according to some embodiments.

DETAILED DESCRIPTIONS

(15) It will be understood that, although the terms “first”, “second”, “third”, and so on may be used herein for illustrating various elements, components, regions, layers and/or sections, these elements, components, regions, layers and/or sections should not be limited by these terms. These terms are used to distinguish one element, component, region, layer or section from another element, component, region, layer or section. Thus, a first element, component, region, layer or section described below could be termed a second element, component, region, layer or section, without departing from the spirit and scope of the present disclosure.

(16) Hereinafter, a semiconductor memory device according to some embodiments will be described with reference to FIGS. **1** to **12**.

(17) FIG. **1** is a circuit diagram for illustrating a semiconductor memory device according to some embodiments. FIG. **2** is an exemplary layout diagram for illustrating the semiconductor memory device of FIG. **1**. FIG. **3** is a cross-sectional view taken along a line A-A of FIG. **2**. FIG. **4** is a cross-sectional view taken along a line B-B in FIG. **2**. FIG. **5** is a cross-sectional view taken along a line C-C in FIG. **2**.

(18) Referring to FIG. **1**, the semiconductor memory device according to some embodiments includes a first unit element I and a second unit element II adjacent to each other.
(19) Each of the first unit element I and the second unit element II may include a pair of inverters INV**1** and INV**2** connected in parallel with each other and connected to and between a power node V.sub.DD and a ground node V.sub.SS. Each of the first unit element I and the second unit element II may also include a first pass transistor PS**1** and a second pass transistor PS**2** respectively connected to output nodes of the inverters INV**1** and INV**2**.
(20) The first pass transistor PS**1** may be connected to a bit-line BL, and the second pass transistor PS**2** may be connected to a complementary bit-line /BL. A gate of each of the first pass transistor PS**1** and the second pass transistor PS**2** may be connected to a word-line WL.
(21) To configure one latch circuit, an input node of the first inverter INV**1** may be connected to an output node of the second inverter INV**2**, while an input node of the second inverter INV**2** is connected to an output node of the first inverter INV**1**.
(22) The first inverter INV**1** may include a first pull-up transistor PU**1** and a first pull-down transistor PD**1** connected in series with each other. The second inverter INV**2** may include a second pull-up transistor PU**2** and a second pull-down transistor PD**2** connected in series with each other. Each of the first pull-up transistor PU**1** and the second pull-up transistor PU**2** may be embodied as a P-type field effect transistor (PFET), and each of the first pull-down transistor PD**1** and the second pull-down transistor PD**2** may be embodied as N-type field effect transistor (NFET).
(23) In some embodiments, the first unit element I and the second unit element II may share one bit-line BL. For example, two complementary bit-lines /BL may extend in parallel to each other and may be around one bit-line BL. In this connection, the first unit element I may be defined between one complementary bit-line /BL of the two complementary bit-lines /BL and the bit-line BL. The second unit element II may be defined between the other complementary bit-line /BL of the two complementary bit-lines /BL and the bit-line BL.
(24) Referring to FIG. **2** to FIG. **5**, the semiconductor memory device according to some embodiments may include a substrate **100**, wire patterns **110**A, **110**B, **210**A, and **210**B, a field insulating film **102**, first to fourth gate lines PG**11**, IG**11**, IG**12**, and PG**12**, fifth to eighth gate lines PG**21**, IG**21**, IG**22**, and PG**22**, source/drain areas **160**A and **160**B, a separating insulating film **170**, interlayer insulating films **310** and **320**, a first overlapping contact **190**A and a second overlapping contact **190**B.
(25) The wire patterns **110**A, **110**B, **210**A and **210**B and the first to fourth gate lines PG**11**, IG**11**, IG**12**, and PG**12** may constitute the first unit element I. The wire patterns **110**A, **110**B, **210**A and **210**B and the fifth to eighth gate lines PG**21**, IG**21**, IG**22**, and PG**22** may constitute the second unit element II. Hereinafter, the semiconductor memory device according to some embodiments will be described based on the first unit element I. However, those of ordinary skill in the art to which the present disclosure pertains will understand that the description of the first unit element I may be equally applied to the second unit element II. For example, the fifth to eighth gate lines PG**21**, IG**21**, IG**22**, and PG**22** may correspond to the first to fourth gate lines PG**11**, IG**11**, IG**12**, and PG**12**, respectively. In some embodiments, the first unit element I and the second unit element II may be symmetrically arranged with each other around a boundary between the first unit element I and the second unit element II in a plan view.
(26) The substrate **100** may be made of bulk silicon or SOI (silicon-on-insulator). Alternatively, the substrate **100** may be embodied as a silicon substrate, or may be made of a material other than silicon, such as silicon germanium, SGOI (silicon germanium on insulator), indium antimonide, lead telluride, indium arsenide, indium phosphide, gallium arsenide or gallium antimonide, with the understanding that the present disclosure is not limited to these examples. Alternatively, the substrate **100** may include a base substrate and an epitaxial layer formed on the base substrate. For convenience of description, an example in which the substrate **100** is embodied as the silicon substrate is described below.

(27) A first lower wire pattern **110**A and a first upper wire pattern **110**B may be sequentially stacked on the substrate **100**. The first lower wire pattern **110**A may be spaced apart from the substrate **100** and may be above the substrate **100**, and the first upper wire pattern **110**B may be spaced apart from the first lower wire pattern **110**A and may be above the first lower wire pattern **110**A. Each of the first lower wire pattern **110**A and the first upper wire pattern **110**B may extend in a first direction (e.g., a Y direction) parallel to a top surface of the substrate **100**.
(28) In some embodiments, the first lower wire pattern **110**A may include a plurality of nanosheets (e.g., first to third nanosheets **112**, **113**, and **114**) sequentially stacked on the substrate **100** and spaced apart from each other. In some embodiments, the first upper wire pattern **110**B may include a plurality of nanosheets (e.g., fourth to sixth nanosheets **115**, **116**, and **117**) that are sequentially stacked on the first lower wire pattern **110**A and spaced apart from each other.
(29) In some embodiments, a first pin-like pattern **111** may be formed between the substrate **100** and the first lower wire pattern **110**A. The first pin-like pattern **111** may protrude from the top surface of the substrate **100** and extend in the first direction (e.g., the Y direction). The first pin-like pattern **111** may be formed by etching a portion of the substrate **100**, or may be embodied as an epitaxial layer grown from the substrate **100**.
(30) The second lower wire pattern **210**A and the second upper wire pattern **210**B may be sequentially stacked on the substrate **100**. The second lower wire pattern **210**A may be spaced apart from the substrate **100** and may be above the substrate **100**, and the second upper wire pattern **210**B may be spaced apart from the second lower wire pattern **210**A and may be above the second lower wire pattern **210**A. Each of the second lower wire pattern **210**A and the second upper wire pattern **210**B may extend in the first direction (e.g., the Y direction). The second lower wire pattern **210**A and the second upper wire pattern **210**B may be spaced apart from the first lower wire pattern **110**A and the first upper wire pattern **110**B in a second direction (e.g., the X direction) that is parallel to the top surface of the substrate **100** and that intersects the first direction.
(31) In some embodiments, the second lower wire pattern **210**A may be at the same level as that of the first lower wire pattern **110**A, and the second upper wire pattern **210**B may be at the same level as that of the first upper wire pattern **110**B. As used herein, layers that are “at the same layer” as each other (e.g., “a first layer is at the same level as a second layer”) means that a vertical dimension between the first layer and the top surface of the substrate **100** is equal to a vertical dimension between the second layer and the top surface of the substrate **100**. Further, as used herein, the terms “the same” and “equal” are intended to encompass not only completely the same, but also substantially the same which includes an insignificant difference that may occur due to a margin on a process.
(32) In some embodiments, the second lower wire pattern **210**A may include a plurality of nanosheets (e.g., seventh to ninth nanosheets **212**, **213**, and **214**) that are sequentially stacked on the substrate **100** and spaced apart from each other. In some embodiments, the second upper wire pattern **210**B may include a plurality of nanosheets (e.g., tenth to twelfth nanosheets **215**, **216**, and **217**) that are sequentially stacked on the second lower wire pattern **210**A and spaced apart from each other.
(33) In some embodiments, a second pin-like pattern **211** may be formed between the substrate **100** and the second lower wire pattern **210**A. The second pin-like pattern **211** may protrude from the top surface of the substrate **100** and extend in the first direction (e.g., the Y direction). The second pin-like pattern **211** may be formed by etching a portion of the substrate **100**, or may be embodied as an epitaxial layer grown from the substrate **100**.
(34) Each of the wire patterns **110**A, **110**B, **210**A and **210**B may include an elemental semiconductor material such as silicon (Si) or germanium (Ge). Alternatively, each of the wire patterns **110**A, **110**B, **210**A, and **210**B may include a compound semiconductor, for example, a group IV-IV compound semiconductor or a group III-V compound semiconductor. The group IV-IV compound semiconductor may include, and as non-limiting examples, a binary compound

including two of carbon (C), silicon (Si), germanium (Ge), and tin (Sn), a ternary compound including three thereof, or a compound obtained by doping a group IV element thereto. The group III-V compound semiconductor may include, and as non-limiting examples, a binary compound obtained by combining one of aluminum (Al), gallium (Ga), and indium (In) as a group III element and one of phosphorus (P), arsenic (As), and antimony (Sb) as a group V element with each other, a ternary compound obtained by combining two of aluminum (Al), gallium (Ga), and indium (In) as a group III element and one of phosphorus (P), arsenic (As), and antimony (Sb) as a group V with each other, or a quaternary compound obtained by combining three of aluminum (Al), gallium (Ga), and indium (In) as a group III element and one of phosphorus (P), arsenic (As), and antimony (Sb) as a group V with each other.
(35) As best seen in FIG. **5**, the field insulating film **102** may be formed on the substrate **100**. In some embodiments, the field insulating film **102** may cover at least a portion of a side surface of the first pin-like pattern **111** and at least a portion of a side surface of the second pin-like pattern **211**. The field insulating film **102** may include, for example, at least one of an oxide film, a nitride film, an oxynitride film, and a combination thereof, with the understanding that the present disclosure is not limited thereto.
(36) The first gate line PG**11** may extend in the second direction (e.g., the X direction) and may intersect the first lower wire pattern **110**A and the first upper wire pattern **110**B. The second gate line IG**11** may be spaced apart from the first gate line PG**11** in the first direction (e.g., the Y direction). The second gate line IG**11** may extend in the second direction (e.g., the X direction) and may intersect the first lower wire pattern **110**A and the first upper wire pattern **110**B.
(37) Each of the first gate line PG**11** and the second gate line IG**11** may surround a side surface of the first lower wire pattern **110**A and a side surface of the first upper wire pattern **110**B. That is, each of the first lower wire pattern **110**A and the first upper wire pattern **110**B may extend in the first direction (e.g., the Y direction) and may extend through the first gate line PG**11** and the second gate line IG**11**.
(38) The third gate line IG**12** may be spaced apart from the first gate line PG**11** in the second direction X. The third gate line IG**12** may extend in the second direction (e.g., the X direction) and may intersect the second lower wire pattern **210**A and the second upper wire pattern **210**B. The fourth gate line PG**12** may be spaced apart from the second gate line IG**11** in the second direction X. The fourth gate line PG**12** may extend in the second direction (e.g., the X direction) and may intersect the second lower wire pattern **210**A and the second upper wire pattern **210**B.
(39) Each of the third gate line IG**12** and the fourth gate line PG**12** may surround a side surface of the second lower wire pattern **210**A and a side surface of the second upper wire pattern **210**B. That is, each of the second lower wire pattern **210**A and the second upper wire pattern **210**B may extend in the first direction (e.g., the Y direction) and may extend through the third gate line IG**12** and the fourth gate line PG**12**.
(40) Each of the first to fourth gate lines PG**11**, IG**11**, IG**12**, and PG**12** may include a gate dielectric film **120**, a gate electrode **130**, a gate spacer **140**, and a gate capping pattern **150**.
(41) The gate electrode **130** may include, as examples, at least one of TiN, WN, TaN, Ru, TiC, TaC, Ti, Ag, Al, TiAl, TiAlN, TiAlC, TaCN, TaSiN, Mn, Zr, W, Al, and combinations thereof, with the understanding that the present disclosure is not limited thereto. The gate electrode **130** may be formed using a replacement process, with the understanding that the present disclosure is not limited thereto.
(42) Although only a single layer is illustrated as the gate electrode **130**, this is only an example. In some embodiments, the gate electrode **130** may be formed by stacking a plurality of conductive layers. For example, the gate electrode **130** may include a work function adjusting film that may control a work function, and a filling conductive film that fills or is within a space defined by the work function adjusting film. The work function adjusting film may include, for example, at least one of TiN, TaN, TiC, TaC, TiAlC, and/or combinations thereof. The filling conductive film may

include, for example, W or Al.
(43) The gate dielectric film **120** may be interposed between each of the wire patterns **110**A, **110**B, **210**A and **210**B and the gate electrode **130**. Further, the gate dielectric film **120** may be interposed between the first pin-like pattern **111** and the gate electrode **130** and between the second pin-like pattern **211** and the gate electrode **130**. The gate dielectric film **120** may extend along a top surface of the field insulating film **102**.
(44) The gate dielectric film **120** may include, for example, at least one of silicon oxide, silicon oxynitride, silicon nitride, or a high dielectric constant material having a higher dielectric constant than that of silicon oxide. The high dielectric constant material may include, for example, at least one of hafnium oxide, hafnium silicon oxide, hafnium aluminum oxide, lanthanum oxide, lanthanum aluminum oxide, zirconium oxide, zirconium silicon oxide, tantalum oxide, titanium oxide, barium strontium titanium oxide, barium titanium oxide, strontium titanium oxide, yttrium oxide, aluminum oxide, lead scandium tantalum oxide, lead zinc niobate, and/or combinations thereof, with the understanding that the present disclosure is not limited thereto.
(45) Although not shown, in some embodiments an interface film may be formed between each of the wire patterns **110**A, **110**B, **210**A, and **210**B and the gate dielectric film **120**. The interfacial film may include, for example, an oxide film, although the present disclosure is not limited thereto.
(46) The gate spacer **140** may be formed on the substrate **100** and the field insulating film **102**. The gate spacer **140** may extend along a side surface of the gate electrode **130**. In some embodiments, the gate dielectric film **120** may further extend along an inner side surface of the gate spacer **140**. For example, the gate dielectric film **120** may be interposed between the gate electrode **130** and the gate spacer **140**. The gate dielectric film **120** may be formed using a replacement process, although the present disclosure is not limited thereto.
(47) The gate spacer **140** may include, for example, at least one of silicon nitride, silicon oxynitride, silicon oxycarbide, silicon boron nitride, silicon boron carbonitride, silicon oxycarbonitride, and combinations thereof, with the understanding that the present disclosure is not limited thereto.
(48) In some embodiments, a first inner spacer **145**A may be formed on a side surface of the gate electrode **130** between adjacent ones of the first to third nanosheets **112**, **113**, and **114**. The first inner spacer **145**A may also be formed between the first pin-like pattern **111** and the first nanosheet **112**.
(49) The first inner spacer **145**A may include, as non-limiting examples, at least one of silicon nitride, silicon oxynitride, silicon oxycarbide, silicon boron nitride, silicon boron carbonitride, silicon oxycarbonitride, and/or combinations thereof. The first inner spacer **145**A may include the same material as that of the gate spacer **140**, or may include a material different from that of the gate spacer **140**.
(50) In some embodiments, a second inner spacer **145**B may be formed on a side surface of the gate electrode **130** between adjacent ones of the fourth to sixth nanosheets **115**, **116**, and **117**. The second inner spacer **145**B may include, as non-limiting examples, at least one of silicon nitride, silicon oxynitride, silicon oxycarbide, silicon boron nitride, silicon boron carbonitride, silicon oxycarbonitride, and combinations thereof. The second inner spacer **145**B may include the same material as that of each of the first inner spacer **145**A and/or gate spacer **140**, or may include a material different from that of each of the first inner spacer **145**A and/or the gate spacer **140**.
(51) The gate capping pattern **150** may extend along a top surface of the gate electrode **130**. The gate capping pattern **150** may cover at least a portion of the top surface of the gate electrode **130**. Although it is shown in the drawing that a top surface of the gate spacer **140** is coplanar with a top surface of the gate capping pattern **150**, this is merely one example. In some embodiments, the gate capping pattern **150** may be formed to cover the top surface of the gate spacer **140**. The gate capping pattern **150** may include, as non-limiting examples, at least one of silicon nitride, silicon oxynitride, silicon oxycarbide, silicon boron nitride, silicon boron carbonitride, silicon

oxycarbonitride, and/or combinations thereof. The present disclosure is not limited thereto.
(52) The source/drain areas **160**A and **160**B may be formed in the wire patterns **110**A, **110**B, **210**A and **210**B. Hereinafter, a manner in which the source/drain areas **160**A and **160**B may be formed in the first lower wire pattern **110**A and the first upper wire pattern **110**B will be described. However, those of ordinary skill in the art to which the present disclosure pertains will understand that the source/drain areas **160**A and **160**B may also be formed in the second lower wire pattern **210**A and the second upper wire pattern **210**B similarly to the manner described herein.
(53) The source/drain areas **160**A and **160**B may be formed on side surfaces of the first to fourth gate lines PG**11**, IG**11**, IG**12**, and PG**12** and on side surfaces of the fifth to eighth gate lines PG**21**, IG**21**, IG**22**, and PG**22**. The source/drain areas **160**A and **160**B may be electrically separated from the gate electrode **130** via the gate spacer **140**.
(54) The source/drain areas **160**A and **160**B may include a lower source/drain area **160**A formed in the first lower wire pattern **110**A and an upper source/drain area **160**B formed in the first upper wire pattern **110**B. The lower source/drain area **160**A may be connected to the first lower wire pattern **110**A, and the upper source/drain area **160**B may be connected to the first upper wire pattern **110**B.
(55) Each of the source/drain areas **160**A and **160**B may include an epitaxial layer. For example, each of the source/drain areas **160**A and **160**B may be formed using an epitaxial growth method. In some embodiments, a cross section of each of the source/drain areas **160**A and **160**B intersecting the second direction X may have a diamond shape (or a pentagonal shape, or a hexagonal shape).
(56) The lower source/drain area **160**A may have a first conductivity type, and the upper source/drain area **160**B may have a second conductivity type different from that of the lower source/drain area **160**A. In some examples of embodiments, the first conductivity type may be a n-type and the second conductivity type may be a p-type. In this case, the first lower wire pattern **110**A and the second lower wire pattern **210**A may be used as a channel area of NFET, and the first upper wire pattern **110**B and the second upper wire pattern **210**B may be used as a channel area of PFET. However, this is only one example, and in some embodiments the first conductivity type may be a p-type and the second conductivity type may be a n-type.
(57) When a semiconductor memory device including the first lower wire pattern **110**A and the second lower wire pattern **210**A is NFET, the lower source/drain area **160**A may contain an n-type impurity or an impurity for preventing diffusion of the n-type impurity. For example, the lower source/drain area **160**A may contain at least one of P, Sb, As, and/or combinations thereof.
(58) In some embodiments, the lower source/drain area **160**A may include a tensile stress material. For example, when each of the first lower wire pattern **110**A and the second lower wire pattern **210**A includes silicon (Si), the lower source/drain area **160**A may include a material (e.g., SiC) that has a lattice constant smaller than that of silicon (Si). The tensile stress material may apply tensile stress to the first lower wire pattern **110**A and the second lower wire pattern **210**A to improve carrier mobility in the channel area.
(59) When the semiconductor memory device including the first upper wire pattern **110**B and the second upper wire pattern **210**B is PFET, the upper source/drain area **160**B may contain a p-type impurity or an impurity for preventing diffusion of the p-type impurity. For example, the lower source/drain area **160**A may contain at least one of B, C, In, Ga, Al, and/or combinations thereof.
(60) In some embodiments, the upper source/drain area **160**B may include a compressive stress material. For example, when each of the first upper wire pattern **110**B and the second upper wire pattern **210**B includes silicon (Si), the upper source/drain area **160**B may include a material having a lattice constant larger than that of silicon (Si). For example, the upper source/drain area **160**B may include silicon germanium (SiGe). The compressive stress material may apply compressive stress to the first lower wire pattern **110**A and the second lower wire pattern **210**A to improve carrier mobility in the channel area.
(61) The separating insulating film **170** may be formed on the substrate **100** and the field insulating

film **102**. The separating insulating film **170** may cover the lower source/drain area **160**A. The upper source/drain area **160**B may be formed on the separating insulating film **170**. The separating insulating film **170** may include a first separating portion **172** and a second separating portion **174**.
(62) The first separating portion **172** may electrically separate the lower source/drain area **160**A and the upper source/drain area **160**B from each other. For example, the first separating portion **172** may be between the first gate line PG**11** and the second gate line IG**11**, and may electrically separate the lower source/drain area **160**A and the upper source/drain area **160**B between the first gate line PG**11** and the second gate line IG**11** from each other.
(63) The second separating portion **174** may separate the first unit element I and the second unit element II from each other. For example, the second separating portion **174** may be interposed between the first gate line PG**11** and the fifth gate line PG**21** (and between the third gate line IG**12** and the seventh gate line IG**22**) to separate the first unit element I and the second unit element II from each other.
(64) In some embodiments, the first separating portion **172** and the second separating portion **174** may be monolithic with each other. That is, the first separating portion **172** and the second separating portion **174** may be formed using the same manufacturing process and may have the same material composition.
(65) In some embodiments, a vertical level of a top surface of the second separating portion **174** may be higher than that of a top surface of the first separating portion **172**. For example, as shown in FIG. **3**, a vertical dimension H**1** of the second separating portion **174** (as measured from an upper surface of the lower source/drain area **160**A to an upper surface of the second separating portion **174**) may be larger than a vertical dimension H**2** of the first separating portion **172** (as measured from the upper surface of the lower source/drain area **160**A to an upper surface of the first separating portion **172**).
(66) In some embodiments, a vertical level of the top or upper surface of the second separating portion **174** may be higher than or equal to that of a top surface of the first upper wire pattern **110**B (and a top surface of the second upper wire pattern **210**B). In this case, the upper source/drain area **160**B may not be interposed between the first gate line PG**11** and the fifth gate line PG**21** (and between the third gate line IG**12** and the seventh gate line IG**22**).
(67) The separating insulating film **170** may include, for example, at least one of silicon oxide, silicon nitride, silicon oxynitride, silicon oxycarbide, silicon boron nitride, silicon boron carbonitride, silicon oxycarbonitride, and/or a low dielectric constant material. The low dielectric constant material may include, for example, at least one of FOX (Flowable Oxide), TOSZ (Torene SilaZene), USG (Undoped Silica Glass), BSG (Borosilica Glass), PSG (PhosphoSilica Glass), BPSG (BoroPhosphoSilica Glass), PETEOS (Plasma Enhanced Tetra Ethyl Ortho Silicate), FSG (Fluoride Silicate Glass), CDO (Carbon Doped silicon Oxide), Xerogel, Aerogel, Amorphous Fluorinated Carbon, OSG (Organo Silicate Glass), Parylene, BCB (bis-benzocyclobutenes), SiLK, polyimide, a porous polymeric material, and/or combinations thereof, with the understanding that the present disclosure is not limited to these examples.
(68) The interlayer insulating films **310** and **320** may cover the first to fourth gate lines PG**11**, IG**11**, IG**12**, and PG**12**, the fifth to eighth gate lines PG**21**, IG**21**, IG**22**, and PG**22**, the source/drain areas **160**A and **160**B and the separating insulating film **170**. Each of the interlayer insulating films **310** and **320** may include, for example, at least one of silicon oxide, silicon oxynitride, and/or a low dielectric constant material. In some embodiments, the interlayer insulating films **310** and **320** may include a first interlayer insulating film **310** and a second interlayer insulating film **320** that are sequentially stacked.
(69) The first interlayer insulating film **310** may cover a top surface of the separating insulating film **170**, a side surface of each of the first to fourth gate lines PG**11**, IG**11**, IG**12**, and PG**12**, a side surface of each of the fifth to eighth gate lines PG**21**, IG**21**, IG**22**, and PG**22**, and a top surface of the upper source/drain area **160**B. In some embodiments, a top surface of the first interlayer

insulating film **310** may be coplanar with a top surface of each of the first to fourth gate lines PG**11**, IG**11**, IG**12**, and PG**12** and a top surface of each of the fifth to eighth gate lines PG**21**, IG**21**, IG**22**, and PG**22**.
(70) The second interlayer insulating film **320** may cover a top surface of the first interlayer insulating film **310**, a top surface of each of the first to fourth gate lines PG**11**, IG**11**, IG**12**, and PG**12**, and a top surface of each of the fifth to eighth gate lines PG**21**, IG**21**, IG**22**, and PG**22**.
(71) The first gate line PG**11** and/or the fourth gate line PG**12** may further include a recess capping pattern **155**. Hereinafter, a manner in which the recess capping pattern **155** of the first gate line PG**11** is formed is described. However, those of ordinary skill in the art to which the present disclosure pertains will understand that the recess capping pattern **155** of the fourth gate line PG**12** may also be formed similarly to the manner described herein.
(72) The recess capping pattern **155** may cover at least a portion of a top surface of the gate electrode **130** that overlaps the first overlapping contact **190**A which will be described later. Herein, the term "overlapping" or "overlap" may refer to overlapping in the third direction Z that intersects the top surface of the substrate **100**. For example, the recess capping pattern **155** may extend along a portion of a top surface of the gate electrode **130** of the first gate line PG**11** around the first upper wire pattern **110**B. In some embodiments, the gate capping pattern **150** of the first gate line PG**11** may extend along a portion of a top surface of the gate electrode **130** of the first gate line PG**11** where the recess capping pattern **155** is not formed.
(73) In some embodiments, the top surface of the recess capping pattern **155** may be coplanar with the top surface of the gate capping pattern **150**. For example, both the top surface of the recess capping pattern **155** and the top surface of the gate capping pattern **150** may be coplanar with the top surface of the first interlayer insulating film **310**.
(74) A vertical level of a bottom surface of the recess capping pattern **155** may be lower than that of a bottom surface of the gate capping pattern **150**. For example, as shown in FIG. **5**, a depth D**2** of the bottom surface of the recess capping pattern **155** from the top surface of the first interlayer insulating film **310** may be greater than a depth D**1** of the bottom surface of the gate capping pattern **150** from the top surface of the first interlayer insulating film **310**.
(75) In some embodiments, the vertical level of the bottom surface of the recess capping pattern **155** may be higher than or equal to that of a top surface of the first upper wire pattern **110**B (and a top surface of the second upper wire pattern **210**B).
(76) In some embodiments, the gate spacer **140** may be interposed between the recess capping pattern **155** and the first interlayer insulating film **310**.
(77) The recess capping pattern **155** may include, for example, at least one of silicon nitride, silicon oxynitride, silicon oxycarbide, silicon boron nitride, silicon boron carbonitride, silicon oxycarbonitride, and/or combinations thereof, with the understanding that the present disclosure is not limited thereto. In some embodiments, the recess capping pattern **155** may include the same material as that of the gate capping pattern **150**.
(78) FIG. **5** shows that no boundary is formed between the recess capping pattern **155** and the gate capping pattern **150**, however the present disclosure is not limited thereto. In some embodiments, a boundary may be present between the recess capping pattern **155** and the gate capping pattern **150**, depending on a formation method of the recess capping pattern **155** and the gate capping pattern **150**.
(79) The first gate line PG**11** may act as a gate electrode of the first pass transistor (PS**1** of FIG. **1**). For example, as shown in FIG. **5**, a first gate contact **182***a* may extend through the second interlayer insulating film **320** and the gate capping pattern **150** and be connected to the gate electrode **130** of the first gate line PG**11**. The first gate contact **182***a* may act as a node of the word-line (WL of FIG. **1**). Thus, the first pass transistor (PS**1** of FIG. **1**) may be connected to the word-line (WL of FIG. **1**).
(80) The first lower wire pattern **110**A that intersects the first gate line PG**11** may act as a channel

area of the first pass transistor (PS**1** of FIG. **1**). For example, as shown in FIG. **4**, a first source/drain contact **181** connected to the lower source/drain area **160**A may be formed between the first gate line PG**11** and the fifth gate line PG**21**. The first source/drain contact **181** may act as a node of the bit-line (BL of FIG. **1**). Thus, the first pass transistor (PS**1** of FIG. **1**) may be connected to the bit-line (BL of FIG. **1**).

(81) In some embodiments, the first source/drain contact **181** may extend in the third direction Z and may extend through the second interlayer insulating film **320**, the first interlayer insulating film **310**, and the second separating portion **174** of the separating insulating film **170**. As described above, and as best seen in FIG. **4**, due to the second separating portion **174**, the upper source/drain area **160**B may not be formed between the first gate line PG**11** and the fifth gate line PG**21**. Thus, the first source/drain contact **181** may be connected to the lower source/drain area **160**A while not being connected to the upper source/drain area **160**B.

(82) The second gate line IG**11** may act as a gate electrode of the first inverter (INV**1** of FIG. **1**). Further, the first upper wire pattern **110**B intersecting the second gate line IG**11** may act as a channel area of the first pull-up transistor (PU**1** of FIG. **1**). For example, as shown in FIG. **3**, a second source/drain contact **184***a* connected to the upper source/drain area **160**B may be formed at one side around the second gate line IG**11**. The second source/drain contact **184***a* may act as the power node (V.sub.DD of FIG. **1**).

(83) Further, the first lower wire pattern **110**A that intersects the second gate line IG**11** may act as a channel area of the first pull-down transistor (PD**1** of FIG. **1**). For example, as shown in FIG. **4**, a third source/drain contact **185***a* connected to the lower source/drain area **160**A may be formed at one side around the second gate line IG**11**. The third source/drain contact **185***a* may act as the ground node (V.sub.SS FIG. **1**).

(84) In some embodiments, a contact insulating film **175** may be formed that electrically separates the third source/drain contact **185***a* from the upper source/drain area **160**B. The contact insulating film **175** may extend along, for example, a side surface of the third source/drain contact **185***a*. Due to the contact insulating film **175**, the third source/drain contact **185***a* may be connected to the lower source/drain area **160**A while not being connected to the upper source/drain area **160**B.

(85) The lower source/drain area **160**A and the upper source/drain area **160**B between the first gate line PG**11** and the second gate line IG**11** may act as the output node of the first inverter (INV**1** of FIG. **1**). For example, as shown in FIG. **3**, a first common contact **183** connecting the lower source/drain area **160**A and the upper source/drain area **160**B to each other may be formed between the first gate line PG**11** and the second gate line IG**11**. Thus, the output node of the first inverter (INV**1** of FIG. **1**) may be connected to the first pass transistor PS**1**. Further, the first inverter (INV**1** of FIG. **1**) may be connected to and between the power node (V.sub.DD of FIG. **1**) and the ground node (V.sub.SS of FIG. **1**).

(86) In some embodiments, the first common contact **183** may extend in the third direction Z and extend through the upper source/drain area **160**B and the first separating portion **172** of the separating insulating film **170**.

(87) The first lower wire pattern **110**A that intersects the first gate line PG**11** may act as a channel area of the first pass transistor (PS**1** of FIG. **1**). For example, as shown in FIG. **4**, the first source/drain contact **181** connected to the lower source/drain area **160**A may be formed between the first gate line PG**11** and the fifth gate line PG**21**. The first source/drain contact **181** may act as a node of the bit-line (BL of FIG. **1**). Thus, the first pass transistor (PS**1** of FIG. **1**) may be connected to the bit-line (BL of FIG. **1**).

(88) The fourth gate line PG**12** may act as a gate electrode of the second pass transistor (PS**2** of FIG. **1**). For example, as shown in FIG. **2**, a second gate contact **182***b* connected to the fourth gate line PG**12** may be formed. The second gate contact **182***b* may act as a node of the word-line (WL of FIG. **1**). Thus, the second pass transistor (PS**2** of FIG. **1**) may be connected to the word-line (WL of FIG. **1**).

(89) The second lower wire pattern **210**A that intersects the fourth gate line PG**12** may act as a channel area of the second pass transistor (PS**2** of FIG. **1**). For example, as shown in FIG. **2**, a fourth source/drain contact **186** connected to a source/drain area (e.g., the lower source/drain area **160**A) of the second lower wire pattern **210**A may be formed at one side around the fourth gate line PG**12**. The fourth source/drain contact **186** may act as a node of the complementary bit-line (/BL of FIG. **1**). Thus, the second pass transistor (PS**2** of FIG. **1**) may be connected to the complementary bit-line (/BL of FIG. **1**). Because the fourth source/drain contact **186** may be similar to the first source/drain contact **181**, detailed description thereof will be omitted herein.
(90) The third gate line IG**12** may act as the gate electrode of the second inverter (INV**2** of FIG. **1**). Further, the second upper wire pattern **210**B that intersects the third gate line IG**12** may act as a channel area of the second pull-up transistor (PU**2** of FIG. **1**). For example, as shown in FIG. **2**, a fifth source/drain contact **184*b*** connected to a source/drain area (e.g., the upper source/drain area **160**B) of the second upper wire pattern **210**B may be formed at one side around the third gate line IG**12**. The fifth source/drain contact **184*b*** may act as the power node (V.sub.DD of FIG. **1**). Because the fifth source/drain contact **184*b*** may be similar to the second source/drain contact **184*a***, detailed description thereof will be omitted herein.
(91) Further, the second lower wire pattern **210**A intersecting the third gate line IG**12** may act as a channel area of the second pull-down transistor (PD**2** of FIG. **1**). For example, as shown in FIG. **2**, a sixth source/drain contact **185*b*** connected to a source/drain area (e.g., the lower source/drain area **160**A) of the second lower wire pattern **210**A may be formed at one side around the third gate line IG**12**. The sixth source/drain contact **185*b*** may act as the ground node (V.sub.SS of FIG. **1**). Because the sixth source/drain contact **185*b*** may be similar to the third source/drain contact **185*a***, detailed description thereof will be omitted herein.
(92) Source/drain areas (e.g., the lower source/drain area **160**A and the upper source/drain area **160**B) between the third gate line IG**12** and the fourth gate line PG**12** may act as an output node of the second inverter (INV**2** of FIG. **1**). For example, as shown in FIG. **2**, a second common contact **187** connecting the source/drain areas (e.g., the lower source/drain area **160**A and the upper source/drain area **160**B) to each other may be formed between the third gate line IG**12** and the fourth gate line PG**12**. Thus, the output node of the second inverter (INV**2** of FIG. **1**) may be connected to the second pass transistor PS**2**. Further, the second inverter (INV**2** of FIG. **1**) may be connected to and between the power node (V.sub.DD of FIG. **1**) and the ground node (V.sub.SS of FIG. **1**).
(93) The first overlapping contact **190**A may electrically connect the source/drain areas **160**A and **160**B between the first gate line PG**11** and the second gate line IG**11** to each other and may be connected to the third gate line IG**12**. For example, the first overlapping contact **190**A may extend through the second interlayer insulating film **320** to connect the first common contact **183** and the gate electrode **130** of the third gate line IG**12** to each other. Thus, the input node of the second inverter (INV**2** of FIG. **1**) may be connected to the output node of the first inverter (INV**1** of FIG. **1**).
(94) At least a portion of the first overlapping contact **190**A may overlap at least a portion of the first gate line PG**11**. For example, as shown in FIGS. **1** and **3**, a portion of the first overlapping contact **190**A may overlap a portion of the gate electrode **130** of the first gate line PG**11** around the first upper wire pattern **110**B.
(95) A vertical level of a bottom surface of the first overlapping contact **190**A may be higher than a vertical level of a bottom surface of the recess capping pattern **155**. Thus, although the first overlapping contact **190**A partially overlaps with the first gate line PG**11**, the contact **190**A may be electrically isolated from the gate electrode **130** of the first gate line PG**11**. Specifically, as shown in FIG. **5**, the first overlapping contact **190**A may be spaced apart from the gate electrode **130** of the first gate line PG**11** via the recess capping pattern **155**.
(96) In some embodiments, a portion of the first overlapping contact **190**A may overlap at least a

portion of the third gate line IG**12**. For example, as shown in FIG. **2**, the portion of the first overlapping contact **190**A may overlap one end of the third gate line IG**12**.
(97) In some embodiments, a vertical level of the bottom surface of the first overlapping contact **190**A may be lower than or equal to that of the bottom surface of the gate capping pattern **150** and higher than that of the bottom surface of the recess capping pattern **155**. Thus, as shown in FIG. **5**, the first overlapping contact **190**A may be connected to the gate electrode **130** of the third gate line IG**12** while not being connected to the gate electrode **130** of the first gate line PG**11**.
(98) In some embodiments, the first overlapping contact **190**A may include a first extension portion **192** that extends in the first direction (e.g., the Y direction) and the second extension portion **194** that extends in the second direction (e.g., the X direction). The first extension portion **192** may extend in the first direction and overlap a portion of the first lower wire pattern **110**A and a portion of the first upper wire pattern **110**B. The first extension portion **192** may be connected to the first common contact **183**. The second extension portion **194** may extend from the first extension portion **192** in the second direction and overlap a portion of the first gate line PG**11** and a portion of the third gate line IG**12**. The second extension portion **194** may be connected to the gate electrode **130** of the third gate line IG**12**. Thus, the first overlapping contact **190**A may electrically connect the first common contact **183** and the third gate line IG**12** to each other.
(99) The second overlapping contact **190**B may electrically connect the source/drain areas (e.g., the lower source/drain area **160**A and the upper source/drain area **160**B) between the third gate line IG**12** and the fourth gate line PG**12** to each other, and may be electrically connected to the second gate line IG**11**. For example, the second overlapping contact **190**B may connect the second common contact **187** and the gate electrode **130** of the second gate line IG**11** to each other. Thus, the input node of the first inverter (INV**1** of FIG. **1**) may be connected to the output node of the second inverter (INV**2** of FIG. **1**).
(100) In some embodiments, the second overlapping contact **190**B may be at the same vertical level as that of the first overlapping contact **190**A. For example, a top surface of the second overlapping contact **190**B may be coplanar with a top surface of the first overlapping contact **190**A.
(101) In some embodiments, at least a portion of the second overlapping contact **190**B may overlap at least a portion of the fourth gate line PG**12**. Because a shape of the second overlapping contact **190**B may be similar to that of the first overlapping contact **190**A, detailed description thereof will be omitted herein.
(102) In some embodiments, the second overlapping contact **190**B may include a third extension portion **197** that extends in the first direction (e.g., the Y direction) and a fourth extension portion **199** that extends in the second direction (e.g., the X direction). The third extension portion **197** may extend in the first direction and overlap a portion of the second lower wire pattern **210**A and a portion of the second upper wire pattern **210**B. The third extension portion **197** may be connected to the second common contact **187**. The fourth extension portion **199** may extend in the second direction from the third extension portion **197** and overlap a portion of the second gate line IG**11** and a portion of the fourth gate line PG**12**. The fourth extension portion **199** may be connected to the gate electrode **130** of the second gate line IG**11**. Thus, the second overlapping contact **190**B may electrically connect the second common contact **187** and the second gate line IG**11** to each other.
(103) In some embodiments, the first overlapping contact **190**A and the second overlapping contact **190**B may be formed symmetrically with each other around a center of the first unit element I in a plan view.
(104) As an integration density of the semiconductor memory devices increases, individual circuit patterns are becoming more miniatured in order to implement a larger number of semiconductor memory devices in the same area. To this end, semiconductor memory devices using a multi-gate transistor are being studied. However, it may be difficult to improve the integration density of such a semiconductor memory device due to complexity of circuit patterns thereof.

(105) The semiconductor memory device according to some embodiments may include the first overlapping contact **190**A that overlaps the first gate line PG**11**, which may improve the integration density of the semiconductor memory device. Specifically, as described above, the first gate line PG**11** may include the recess capping pattern **155** that has a larger depth than that of the gate capping pattern **150**. Due to this recess capping pattern **155**, the first overlapping contact **190**A may be prevented from electrical connection with the first gate line PG**11** even through the contact **190**A overlaps with the first gate line PG**11**. Thus, the semiconductor memory device having an improved integration density may be realized.
(106) FIGS. **6** to **8** are various cross-sectional views for illustrating a semiconductor memory device according to some embodiments. For convenience of description, descriptions of the same components and configurations as those discussed with respect to FIGS. **1** to **5** are only briefly provided or omitted altogether.
(107) Referring to FIG. **6**, in a semiconductor memory device according to some embodiments, a vertical level of a bottom surface of the recess capping pattern **155** may be higher than that of a top surface of the first upper wire pattern **110**B (and a top surface of the second upper wire pattern **210**B).
(108) In this case, a portion of the gate electrode **130** of the first gate line PG**11** may be interposed between the first upper wire pattern **110**B and the recess capping pattern **155**. For example, the portion of the gate electrode **130** of the first gate line PG**11** may extend along a top surface of the sixth nanosheet **117**.
(109) Referring to FIG. **7**, in a semiconductor memory device according to some embodiments, the gate spacer **140** is not interposed between the recess capping pattern **155** and the first interlayer insulating film **310**.
(110) For example, a side surface of the recess capping pattern **155** may contact the first interlayer insulating film **310**. In some embodiments, a lower side surface of the recess capping pattern **155** may contact an upper side surface of the upper source/drain area **160**B.
(111) Referring to FIG. **8**, a semiconductor memory device according to some embodiments may omit the second inner spacer (**145**B in FIGS. **3** and **4**).
(112) The upper source/drain area **160**B may be electrically separated from the gate electrode **130** via the gate dielectric film **120**. In some embodiments, the lower source/drain area **160**A may include n-type impurities, and the upper source/drain area **160**B may include p-type impurities.
(113) FIG. **9** is an exemplary layout diagram for illustrating a semiconductor memory device according to some embodiments. For convenience of description, the descriptions of some components and configurations similar to those described with reference to FIGS. **1** to **5** are only briefly provided or omitted altogether.
(114) Referring to FIG. **9**, in a semiconductor memory device according to some embodiments, the second overlapping contact **190**B may not overlap the fourth gate line PG**12**.
(115) For example, the second overlapping contact **190**B may include a third extension portion **197**, a fourth extension portion **199**, and a connective portion **198**. The third extension portion **197** may extend in the second direction (e.g., the X direction) and be connected to the second common contact **187**. The third extension portion **197** may not overlap the fourth gate line PG**12**. The fourth extension portion **199** may extend in the second direction and may be connected to the second gate line IG**11**. The fourth extension portion **199** may not overlap the fourth gate line PG**12**. The connective portion **198** may extend in the first direction and connect the third extension portion **197** and the fourth extension portion **199** to each other. The connective portion **198** may not overlap the second gate line IG**11** and the fourth gate line PG**12**.
(116) In some embodiments, the second overlapping contact **190**B may not overlap the fourth gate line PG**12**. Thus, the fourth gate line PG**12** may not include the recess capping pattern **155**.
(117) Only a configuration in which the first overlapping contact **190**A overlaps the first gate line PG**11** and the second overlapping contact **190**B does not overlap the fourth gate line PG**12** has

been described. However, in some embodiments, the first overlapping contact **190**A may not overlap the first gate line PG**11** and the second overlapping contact **190**B may overlap the fourth gate line PG**12**.
(118) FIG. **10** is an exemplary layout diagram for illustrating a semiconductor memory device according to some embodiments. FIG. **11** is a cross-sectional view taken along a line D-D of FIG. **10**. FIG. **12** is a cross-sectional view taken along a line E-E of FIG. **10**.
(119) Referring to FIG. **10** to FIG. **12**, a semiconductor memory device according to some embodiments may include a first wiring pattern **390**A.
(120) The first wiring pattern **390**A may electrically connect the first overlapping contact **190**A and the first common contact **183** to each other. For example, the first wiring pattern **390**A may extend in the first direction (e.g., the Y direction) and overlap a portion of the first overlapping contact **190**A and a portion of the first common contact **183**.
(121) In some embodiments, the first wiring pattern **390**A may be at a higher vertical level than that of each of the first overlapping contact **190**A and the first common contact **183**. For example, a third interlayer insulating film **330** may be formed that covers the first overlapping contact **190**A, the first common contact **183** and the second interlayer insulating film **320**. The first wiring pattern **390**A may be formed in the third interlayer insulating film **330** to connect the first overlapping contact **190**A and the first common contact **183** to each other.
(122) In some embodiments, a first via pattern **395***a* and a second via pattern **395***b* may be formed in the third interlayer insulating film **330**. The first via pattern **395***a* may extend through the third interlayer insulating film **330** and connect the first overlapping contact **190**A with the first wiring pattern **390**A. The second via pattern **395***b* may extend through the third interlayer insulating film **330** and connect the first common contact **183** with the first wiring pattern. **390**A. Thus, the first overlapping contact **190**A and the first common contact **183** may be electrically connected to each other.
(123) In some embodiments, a second wiring pattern **390**B may be formed that electrically connects the second overlapping contact **190**B with the second common contact **187**. Because the second wiring pattern **390**B may be similar to the first wiring pattern **390**A, detailed description thereof will be omitted herein.
(124) FIG. **13** to FIG. **25** are diagrams showing intermediate steps for illustrating a method for manufacturing a semiconductor memory device according to some embodiments. For convenience of description, descriptions of some components and configurations described with reference to FIGS. **1** to **5** are briefly provided or omitted altogether.
(125) Referring to FIG. **13**, a lower active pattern **110**AL, an upper active pattern **110**BL, and a preliminary sacrificial pattern **410**L may be formed on a substrate **100**.
(126) The lower active pattern **110**AL and the upper active pattern **110**BL may be sequentially stacked on the substrate **100**. The preliminary sacrificial pattern **410**L may be on the substrate **100** and may be alternately and vertically arranged with the lower active pattern **110**AL and the upper active pattern **110**BL.
(127) Each of the lower active pattern **110**AL, the upper active pattern **110**BL, and the preliminary sacrificial pattern **410**L may be on the substrate **100** and extend in the first direction (e.g., the Y direction). For example, a sacrificial film and an active film that are alternately stacked on the substrate **100** may be formed. A patterning process of patterning the sacrificial film and the active film may be performed.
(128) Each of the lower active pattern **110**AL and the upper active pattern **110**BL may include silicon (Si) or germanium (Ge) as an elemental semiconductor material. Alternatively, each of the lower active pattern **110**AL and the upper active pattern **110**BL may include a compound semiconductor, for example, a group IV-IV compound semiconductor or a group III-V compound semiconductor.
(129) The preliminary sacrificial pattern **410**L may include a material having an etching selectivity

with respect to the lower active pattern **110**AL and the upper active pattern **110**BL. In some embodiments, each of the lower active pattern **110**AL and the upper active pattern **110**BL may include silicon (Si), and the preliminary sacrificial pattern **410**L may include silicon germanium (SiGe).

(130) Referring to FIG. **13** and FIG. **14**, a dummy gate electrode **430**, a dummy gate capping pattern **450**, and the gate spacer **140** may be formed on the substrate **100**.

(131) For example, the dummy gate electrode **430** and the dummy gate capping pattern **450** may be sequentially stacked on the lower active pattern **110**AL, the upper active pattern **110**BL, and the preliminary sacrificial pattern **410**L. A plurality of dummy gate electrodes **430** may be spaced apart from each other. Each of the plurality of dummy gate electrodes **430** may extend in the second direction X. The gate spacer **140** may extend along a side surface of the dummy gate electrode **430** and a side surface of the dummy gate capping pattern **450**.

(132) Subsequently, the lower active pattern **110**AL, the upper active pattern **110**BL, and the preliminary sacrificial pattern **410**L may be patterned using the dummy gate electrode **430** and the gate spacer **140** as an etching mask. Thus, each of the first lower wire pattern **110**A, the first upper wire pattern **110**B, and a sacrificial pattern **410** extending in the first direction Y may be formed. The sacrificial pattern **410** may be alternately and vertically arranged with the first lower wire pattern **110**A and the first upper wire pattern **110**B and on the substrate **100**. Although not specifically shown, the second lower wire pattern **210**A and the second upper wire pattern **210**B may also be formed using the above-described step.

(133) Referring to FIG. **15**, the first inner spacer **145**A and the second inner spacer **145**B may be formed.

(134) For example, a recess process may be performed into a side surface of the sacrificial pattern **410**. Each of the first inner spacer **145**A and the second inner spacer **145**B may be formed within and/or to fill a recessed area of the sacrificial pattern **410**. Thus, the first inner spacer **145**A may be formed on a side surface of a portion of the sacrificial pattern **410** between adjacent ones of the first to third nanosheets **112**, **113**, and **114**. The second inner spacer **145**B may be formed on a side surface of a portion of the sacrificial pattern **410** between adjacent ones of the fourth to sixth nanosheets **115**, **116**, and **117**.

(135) Referring to FIG. **16**, the lower source/drain area **160**A may be formed.

(136) The lower source/drain area **160**A may be formed on the substrate **100**. Further, the lower source/drain area **160**A may be formed on a side surface of each of the dummy gate electrodes **430**. The lower source/drain area **160**A may be connected to the first lower wire pattern **110**A. The first lower source/drain area **160**A may be formed via an epitaxial growth method using the substrate **100** and the first lower wire pattern **110**A as a seed, with the understanding that the present disclosure is not limited thereto.

(137) The lower source/drain area **160**A may have a first conductivity type. In one example, the first conductivity type may be a n-type.

(138) Referring to FIG. **17**, the separating insulating film **170** may be formed.

(139) The separating insulating film **170** may be formed on the substrate **100**. Further, the separating insulating film **170** may cover the lower source/drain area **160**A. The separating insulating film **170** may include, for example, at least one of silicon oxide, silicon oxynitride, and a low dielectric constant material.

(140) Referring to FIG. **18**, a recess process may be performed into a portion of the separating insulating film **170**.

(141) Thus, the separating insulating film **170** that includes the first separating portion **172** and the second separating portion **174** that has a larger vertical dimension than that of the first separating portion **172** may be formed. For example, the vertical dimension H**1** of the second separating portion **174** based on the topmost face of the lower source/drain area **160**A may be greater than the vertical dimension H**2** of the first separating portion **172** based on the topmost face of the lower

source/drain area **160**A. The second separating portion **174** may be formed in an area between the first gate line PG**11** and the fifth gate line PG**21** which will be described later.
(142) Referring to FIG. **19**, the upper source/drain area **160**B may be formed.
(143) The upper source/drain area **160**B may be formed on the separating insulating film **170**. Further, the upper source/drain area **160**B may be formed on a side surface of each of the dummy gate electrodes **430**. The upper source/drain area **160**B may be connected to the first upper wire pattern **110**B. The upper source/drain area **160**B may be formed via an epitaxial growth method using the first upper wire pattern **110**B as a seed, with the understanding that the present disclosure is not limited thereto. Due to the second separating portion **174**, the upper source/drain area **160**B may not be formed in an area between the first gate line PG**11** and the fifth gate line PG**21**, which will be described later.
(144) The upper source/drain area **160**B may have a second conductivity type different from the first conductivity type. In one example, the second conductivity type may be a p-type.
(145) Referring to FIG. **20**, the dummy gate electrode **430**, the dummy gate capping pattern **450**, and the sacrificial pattern **410** may be removed.
(146) For example, an etching process for removing the dummy gate electrode **430**, the dummy gate capping pattern **450**, and the sacrificial pattern **410** may be performed. The etching process may include a wet etching process, as a non-limiting example. Because the sacrificial pattern **410** may have an etch selectivity with respect to the first lower wire pattern **110**A and the first upper wire pattern **110**B, the sacrificial pattern **410** may be selectively removed.
(147) Referring to FIG. **20** and FIG. **21**, the gate dielectric film **120** and the gate electrode **130** may be sequentially formed.
(148) The gate dielectric film **120** may conform to and extend along a profile of an area from which the sacrificial pattern **410** is removed. The gate electrode **130** may be formed on the gate dielectric film **120**. The gate electrode **130** may be formed to fill at least partially the area from which the sacrificial pattern **410** is removed.
(149) Subsequently, the first interlayer insulating film **310** may be formed that covers the upper source/drain area **160**B. In some embodiments, after the first interlayer insulating film **310** has been formed, a planarization process may be performed. The planarization process may include, for example, a chemical mechanical polishing (CMP) process, but the present disclosure is not limited thereto.
(150) Referring to FIG. **22**, a first recess R**1** may be formed into the gate electrode **130**.
(151) For example, a recess process may be performed into a top surface of the gate electrode **130**. In some embodiments, the recess process may selectively etch the gate dielectric film **120** and the gate electrode **130**. Thus, a vertical level of each of a top surface of the gate dielectric film **120** and a top surface of the gate electrode **130** may be lower than that of a top surface of the gate spacer **140**.
(152) Referring to FIG. **23**, a second recess R**2** may be formed into a portion of the gate electrode **130**.
(153) For example, a recess process may be performed into a top surface of the portion of the gate electrode **130**. Thus, the second recess R**2** may be formed such that the second recess R**2** is deeper than the first recess R**1**. For example, a depth D**2** of a bottom surface of the second recess R**2** from a top surface of the first interlayer insulating film **310** may be greater than a depth D**1** of a bottom surface of the first recess R**1** from the top surface of the first interlayer insulating film **310**. The second recess R**2** may be formed in a partial area of the first gate line PG**11** and a partial area of the fifth gate line PG**21**, which will be described later.
(154) Referring to FIG. **23** and FIG. **24**, the gate capping pattern **150** and the recess capping pattern **155** may be formed.
(155) The gate capping pattern **150** may be formed to fill the first recess R**1**, and the recess capping pattern **155** may be formed to fill the second recess R**2**. Thus, the first gate line PG**11**, the second

gate line IG**11**, the fifth gate line PG**21**, and the sixth gate line IG**21** that are spaced apart from each other and extend in the second direction X may be formed. Although not specifically shown, it may be understood the third gate line IG**12**, the fourth gate line PG**12**, the seventh gate line IG**22**, and the eighth gate line PG**22** may also be formed using one or more of the above-described operations.
(156) Each of the gate capping pattern **150** and the recess capping pattern **155** may include, for example, at least one of silicon nitride, silicon oxynitride, silicon oxycarbide, silicon boron nitride, silicon boron carbonitride, silicon oxycarbonitride, and combinations thereof, with the understanding that the present disclosure is not limited thereto.
(157) It has been described that the gate capping pattern **150** and the recess capping pattern **155** are formed at the same time. However, this is only an example. In some embodiments, the recess capping pattern **155** may be formed after the gate capping pattern **150** has been formed. For example, after the first recess R**1** has formed and the gate capping pattern **150** that at least partially fills the first recess R**1** has been formed, the second recess R**2** may be formed and then the recess capping pattern **155** that at least partially fills the second recess R**2** may be formed.
(158) Then, referring to FIG. **2** to FIG. **5**, the first common contact **183**, the second common contact **187**, the first source/drain contact **181**, the second source/drain contact **184***a*, the third source/drain contact **185***a*, the fourth source/drain contact **186**, the fifth source/drain contact **184***b*, the sixth source/drain contact **185***b*, the first overlapping contact **190**A and the second overlapping contact **190**B may be formed. Thus, the semiconductor memory device as described above with reference to FIG. **2** to FIG. **5** may be manufactured.
(159) As described above, the first overlapping contact **190**A may be formed to overlap the first gate line PG**11**. In this connection, the first overlapping contact **190**A may be formed to overlap the recess capping pattern **155** that has the larger depth than that of the gate capping pattern **150**. In this way, methods for manufacturing semiconductor memory devices having improved integration density and reduced process difficulty using simplified processes may be realized.
(160) Further, in methods for manufacturing the semiconductor memory devices according to some embodiments, the first unit element I and the second unit element II may be separated from each other via a simplified process. Specifically, as described above, the first unit element I and the second unit element II may be separated from each other via the second separating portion **174** that is formed using the recess process into the portion of the separating insulating film **170**. Thus, the semiconductor memory device in which the process difficulty is further reduced may be realized.
(161) FIG. **25** is a diagram showing an intermediate step for illustrating a method for manufacturing a semiconductor memory device according to some embodiments. For convenience of description, some of the components and configurations previously described with reference to FIGS. **1** to **14** are only briefly described or discussion thereof is omitted altogether. For reference, FIG. **25** is a diagram of an intermediate step for illustrating the steps after FIG. **22**.
(162) Referring to FIG. **25**, the second recess R**2** may be formed so as not to expose a top surface of the first upper wire pattern **110**B. Subsequently, the steps as described above with reference to FIG. **24** and FIG. **2** to FIG. **5** may be performed. Thus, the semiconductor memory device as described above with reference to FIG. **6** may be manufactured.
(163) FIG. **26** is a diagram of an intermediate step for illustrating a method for manufacturing a semiconductor memory device according to some embodiments. For convenience of description, some of the components and configurations previously described with reference to FIGS. **1** to **24** are only briefly described or discussion thereof is omitted altogether. For reference, FIG. **26** is a diagram of an intermediate step for illustrating the steps after FIG. **22**.
(164) Referring to FIG. **26**, the second recess R**2** may be formed to remove the gate spacer **140**. Subsequently, the steps as described above with reference to FIG. **24** and FIG. **2** to FIG. **5** may be performed. Thus, the semiconductor memory device as described above with reference to FIG. **7** may be manufactured.
(165) FIG. **27** is a diagram of an intermediate step for illustrating a method for manufacturing a

semiconductor memory device according to some embodiments. For convenience of description, some of the components and configurations previously described with reference to FIGS. **1** to **24** are only briefly described or discussion thereof is omitted altogether. For reference, FIG. **27** is a diagram of an intermediate step for illustrating the steps after FIG. **14**.
(166) Referring to FIG. **27**, the first inner spacer **145**A may be formed.
(167) For example, a recess process may be selectively performed on a side surface of a portion of the sacrificial pattern **410** between adjacent ones of the first to third nanosheets **112**, **113**, and **114**. In one example, a portion of the sacrificial pattern **410** between adjacent ones of the first to third nanosheets **112**, **113**, and **114** may have an etching selectivity with respect to a portion of the sacrificial pattern **410** between adjacent ones of the fourth to sixth nanosheets **115**, **116**, and **117**. The first inner spacer **145**A may be formed to fill a recessed area of the sacrificial pattern **410**. In this case, the second inner spacer **145**B as described above using FIG. **15** may not be formed.
(168) Subsequently, the steps as described above with reference to FIG. **16** to FIG. **24** and FIG. **2** to FIG. **5** may be performed. Thus, the semiconductor memory device as described above with reference to FIG. **8** may be manufactured.
(169) While the present inventive concepts have been particularly shown and described with reference to some examples of embodiments thereof, it will be understood by those of ordinary skill in the art that various changes in form and details may be made therein without departing from the scope of the present inventive concepts as defined by the following claims. It is therefore desired that the present examples of embodiments be considered in all respects as illustrative and not restrictive, reference being made to the appended claims rather than the foregoing description to indicate the scope of the present application.

Claims

1. A semiconductor memory device comprising: a substrate; a first lower wire pattern and a first upper wire pattern sequentially stacked on the substrate, and spaced apart from each other, each extending in a first direction; a second lower wire pattern and a second upper wire pattern sequentially stacked on the substrate, and spaced apart from each other, each extending in the first direction, the second lower wire pattern and the second upper wire pattern spaced apart from the first lower wire pattern and the first upper wire pattern in a second direction that intersects the first direction; a first gate line extending in the second direction, and surrounding the first lower wire pattern and the first upper wire pattern; a second gate line extending in the second direction, and surrounding the second lower wire pattern and the second upper wire pattern, the second gate line spaced apart from the first gate line in the second direction; a first lower source/drain area having a first conductivity type, on one side surface of the first gate line, and connected to the first lower wire pattern; a first upper source/drain area having a second conductivity type different from the first conductivity type, on one side surface of the first gate line, and connected to the first upper wire pattern; and a first overlapping contact that electrically connects the first lower source/drain area, the first upper source/drain area and the second gate line to each other, wherein the first overlapping contact at least partially vertically overlaps the first gate line, wherein the first gate line includes a first gate electrode and a recess capping pattern, wherein the recess capping pattern covers a top surface of the first gate electrode that overlaps the first overlapping contact, wherein the second gate line includes a second gate electrode and a gate capping pattern, wherein the gate capping pattern covers a top surface of the second gate electrode, wherein a vertical level of a bottom surface of the recess capping pattern is lower than a vertical level of a bottom surface of the gate capping pattern, wherein a vertical level of a bottom surface of the first overlapping contact is lower than or equal to the vertical level of the bottom surface of the gate capping pattern, and wherein the vertical level of the bottom surface of the first overlapping contact is higher than the vertical level of the bottom surface of the recess capping pattern.

2. The semiconductor memory device of claim 1, wherein a top surface of the recess capping pattern and a top surface of the gate capping pattern are coplanar with each other.
3. The semiconductor memory device of claim 1, wherein the vertical level of the bottom surface of the recess capping pattern is higher than or equal to a vertical level of a top surface of the first upper wire pattern.
4. The semiconductor memory device of claim 1, wherein the first overlapping contact includes: a first extension portion that extends in the first direction and overlaps the first lower wire pattern and the first upper wire pattern; and a second extension portion that extends from the first extension portion in the second direction and overlaps the first gate line and the second gate line.
5. The semiconductor memory device of claim 1, wherein the first conductivity type is a n-type, and the second conductivity type is a p-type.
6. The semiconductor memory device of claim 1, further comprising a separating insulating film that includes: a first separating portion on one side of the first gate line that separates the first lower source/drain area and the first upper source/drain area from each other; and a second separating portion on an opposite side of the first gate line, wherein a vertical level of a top surface of the second separating portion is higher than a vertical level of a top surface of the first separating portion.
7. The semiconductor memory device of claim 6, wherein the vertical level of the top surface of the second separating portion is higher than or equal to a vertical level of a top surface of the first upper wire pattern.
8. The semiconductor memory device of claim 1, further comprising: a third gate line extending in the second direction, and surrounding the first lower wire pattern and the first upper wire pattern, and spaced apart from the first gate line in the first direction; a fourth gate line extending in the second direction, and surrounding the second lower wire pattern and the second upper wire pattern, and spaced apart from the third gate line in the second direction; a second lower source/drain area having the first conductivity type, between the second gate line and the fourth gate line, and connected to the second lower wire pattern; a second upper source/drain area having the second conductivity type, between the second gate line and the fourth gate line, and connected to the second upper wire pattern; and a second overlapping contact that electrically connects the second lower source/drain area, the second upper source/drain area, and the third gate line to each other.
9. A semiconductor memory device comprising: a substrate; a first lower wire pattern and a first upper wire pattern sequentially stacked on the substrate, and spaced apart from each other, each extending in a first direction; a second lower wire pattern and a second upper wire pattern sequentially stacked on the substrate, and spaced apart from each other, each extending in the first direction, the second lower wire pattern and the second upper wire pattern spaced apart from the first lower wire pattern and the first upper wire pattern in a second direction that intersects the first direction; a first gate line extending in the second direction, and surrounding the first lower wire pattern and the first upper wire pattern; a second gate line extending in the second direction, and surrounding the second lower wire pattern and the second upper wire pattern, wherein the second gate line is spaced apart from the first gate line in the second direction; a first lower source/drain area having a first conductivity type, on one side of the first gate line, and connected to the first lower wire pattern; a first upper source/drain area having a second conductivity type different from the first conductivity type, on one side of the first gate line, and connected to the first upper wire pattern; a common contact extending in a third direction that intersects a top surface of the substrate, wherein the common contact is connected to the first lower source/drain area and the first upper source/drain area; and an overlapping contact that electrically connects the common contact and the second gate line to each other, wherein the overlapping contact at least partially overlaps the first gate line, wherein the first gate line includes a first gate electrode and a recess capping pattern, wherein the recess capping pattern covers a top surface of the first gate electrode that overlaps the overlapping contact, wherein the second gate line includes a second gate electrode and

a gate capping pattern, wherein the gate capping pattern covers a top surface of the second gate electrode, wherein a vertical level of a bottom surface of the overlapping contact is lower than or equal to a vertical level of the bottom surface of the gate capping pattern, and wherein the vertical level of the bottom surface of the overlapping contact is higher than a vertical level of the bottom surface of the recess capping pattern.

10. The semiconductor memory device of claim 9, wherein the overlapping contact includes: a first extension portion that extends in the first direction and is connected to the common contact; and a second extension portion that extends from the first extension portion in the second direction and overlaps the first gate line and the second gate line.

11. The semiconductor memory device of claim 9, further comprising a wiring pattern that extends in the first direction and connects the common contact and the overlapping contact to each other, wherein the overlapping contact extends in the second direction and overlaps the first gate line and the second gate line.

12. The semiconductor memory device of claim 11, wherein the wiring pattern is on a top surface of the common contact and a top surface of the overlapping contact.

13. The semiconductor memory device of claim 9, further comprising: a second lower source/drain area having the first conductivity type, on one side of the second gate line, and connected to the second lower wire pattern; a second upper source/drain area having the second conductivity type, on one side of the second gate line, and connected to the second upper wire pattern; a first lower source/drain contact connected to the second lower source/drain area and isolated from the second upper source/drain area; and a first upper source/drain contact connected to the second upper source/drain area and isolated from the second lower source/drain area.

14. A semiconductor memory device comprising: a substrate; a first lower wire pattern and a first upper wire pattern sequentially stacked on the substrate, and spaced apart from each other, each extending in a first direction; a second lower wire pattern and a second upper wire pattern sequentially stacked on the substrate, and spaced apart from each other, each extending in the first direction, the second lower wire pattern and the second upper wire pattern spaced apart from the first lower wire pattern and the first upper wire pattern in a second direction that intersects the first direction; a first gate line extending in the second direction, and surrounding the first lower wire pattern and the first upper wire pattern; a second gate line extending in the second direction, and surrounding the second lower wire pattern and the second upper wire pattern, wherein the second gate line is spaced apart from the first gate line in the second direction; a third gate line extending in the second direction, and surrounding the first lower wire pattern and the first upper wire pattern, and spaced apart from the first gate line in the first direction; a fourth gate line extending in the second direction, and surrounding the second lower wire pattern and the second upper wire pattern, and spaced apart from the third gate line in the second direction; a first lower source/drain area having a first conductivity type, between the first gate line and the third gate line, and connected to the first lower wire pattern; a first upper source/drain area having a second conductivity type different from the first conductivity type, between the first gate line and the third gate line, and connected to the first upper wire pattern; and a first overlapping contact that electrically connects the first lower source/drain area, the first upper source/drain area, and the second gate line to each other, wherein the first overlapping contact at least partially overlaps the first gate line, wherein each of the first to fourth gate lines includes a gate electrode and a gate capping pattern that covers a top surface of the gate electrode, wherein the first gate line further includes a first recess capping pattern that covers a top surface of the gate electrode of the first gate line overlapping the first overlapping contact, and wherein a vertical level of a bottom surface of the first recess capping pattern is lower than a vertical level of a bottom surface of the gate capping pattern wherein a vertical level of a bottom surface of the first overlapping contact is lower than or equal to the vertical level of the bottom surface of the gate capping pattern, and wherein the vertical level of the bottom surface of the first overlapping contact is higher than the vertical level of the bottom surface

of the first recess capping pattern, wherein a vertical level of a bottom surface of the first overlapping contact is lower than or equal to the vertical level of the bottom surface of the gate capping pattern, and wherein the vertical level of the bottom surface of the first overlapping contact is higher than the vertical level of the bottom surface of the first recess capping pattern.

15. The semiconductor memory device of claim 14, further comprising: a second lower source/drain area having the first conductivity type, between the second gate line and the fourth gate line, and connected to the second lower wire pattern; a second upper source/drain area having the second conductivity type, between the second gate line and the fourth gate line, and connected to the second upper wire pattern; and a second overlapping contact that electrically connects the second lower source/drain area, the second upper source/drain area and the third gate line to each other, wherein the first overlapping contact and the second overlapping contact are at the same vertical level.

16. The semiconductor memory device of claim 15, wherein the second overlapping contact at least partially overlaps the fourth gate line, wherein the fourth gate line further includes a second recess capping pattern that covers a top surface of the gate electrode of the fourth gate line that overlaps the second overlapping contact, wherein a vertical level of a bottom surface of the second recess capping pattern is lower than the vertical level of the bottom surface of the gate capping pattern.

17. The semiconductor memory device of claim 14, further comprising: a fifth gate line extending in the second direction, and surrounding the first lower wire pattern and the first upper wire pattern, wherein the fifth gate line is spaced apart from the third gate line with the first gate line therebetween; and a separating insulating film including: a first separating portion between the first gate line and the third gate line that separates the first lower source/drain area and the first upper source/drain area from each other; and a second separating portion between the first gate line and the fifth gate line, wherein a vertical level of a top surface of the second separating portion is higher than a vertical level of a top surface of the first separating portion.

18. The semiconductor memory device of claim 17, wherein the vertical level of the top surface of the second separating portion is higher than or equal to a vertical level of a top surface of the first upper wire pattern.

19. The semiconductor memory device of claim 14, wherein the first conductivity type is a n-type and the second conductivity type is a p-type.